WildlifeReID-10k Model Comparison

Final evaluation of ConvNeXt Base and MiewID on a shared open-set animal re-identification protocol

MiewID is the recommended model: it achieved the strongest Top-1 accuracy and open-set AUROC on the same 606 test queries.

Known identities	Known queries	Unknown queries	Total queries
100	238	368	606

Evaluation scope

The experiment uses the existing trained checkpoints. It does not retrain either model. Both checkpoints contain the same ordered set of 100 identities selected from the WildlifeReID-10k training split with random_state=42.

Evaluation protocol

Known queries are test-split images of the selected training identities. Unknown queries come from 20 identities that never appear in the complete training split. Top-1 and Top-5 use the 238 known queries. Open-set AUROC uses all 606 queries.

Different score types

ConvNeXt produces softmax class probabilities. MiewID generates an embedding and retrieves identities using cosine similarity. These scores do not share a common scale, so the comparison uses threshold-independent AUROC rather than treating the same numeric confidence threshold as equivalent for both models.

Results

Model	Approach	Top-1	Top-5	AUROC	Mean time
ConvNeXt Base	Softmax classification	82.77%	97.90%	90.81%	34.1 ms
MiewID	Cosine retrieval	88.24%	95.80%	95.86%	91.2 ms

MiewID leads Top-1 by 5.47 percentage points and open-set AUROC by 5.05 percentage points. ConvNeXt leads Top-5 by 2.10 percentage points and has the lower mean inference time.

Identification performance

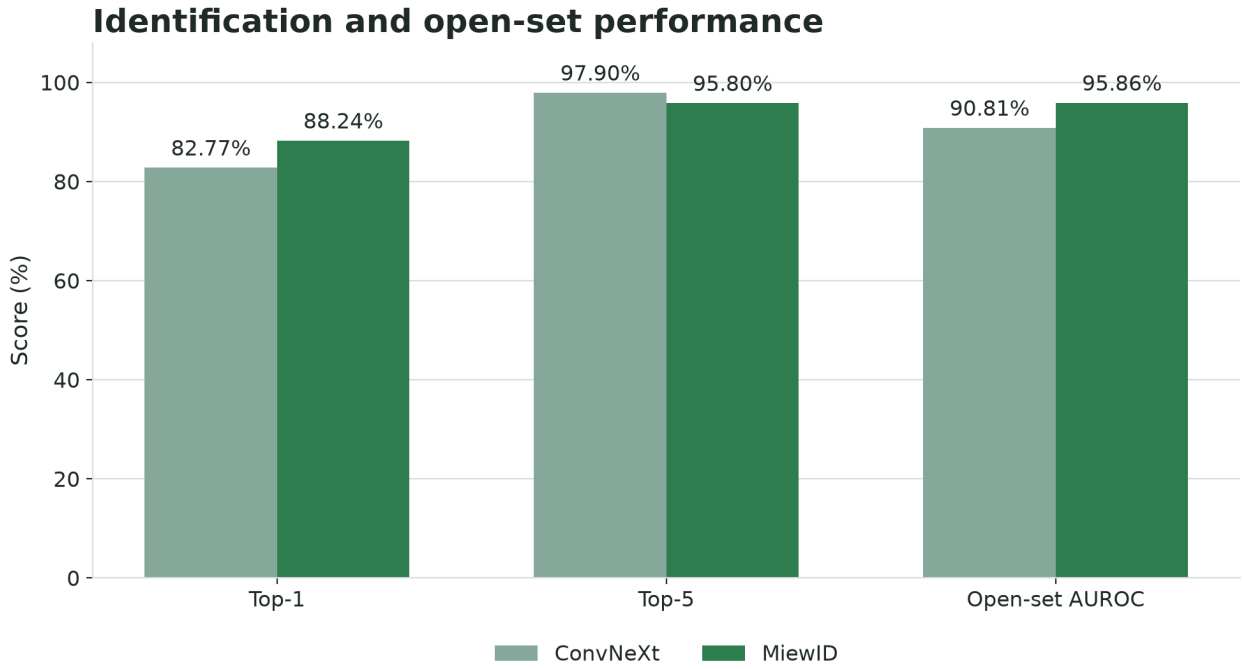


Figure 1. Closed-set identification metrics use known queries only; open-set AUROC uses known and unknown queries.

MiewID makes the strongest first-choice predictions and separates unseen animals more reliably. ConvNeXt's high Top-5 score shows that the correct identity often remains among its leading candidates.

Unknown-animal detection

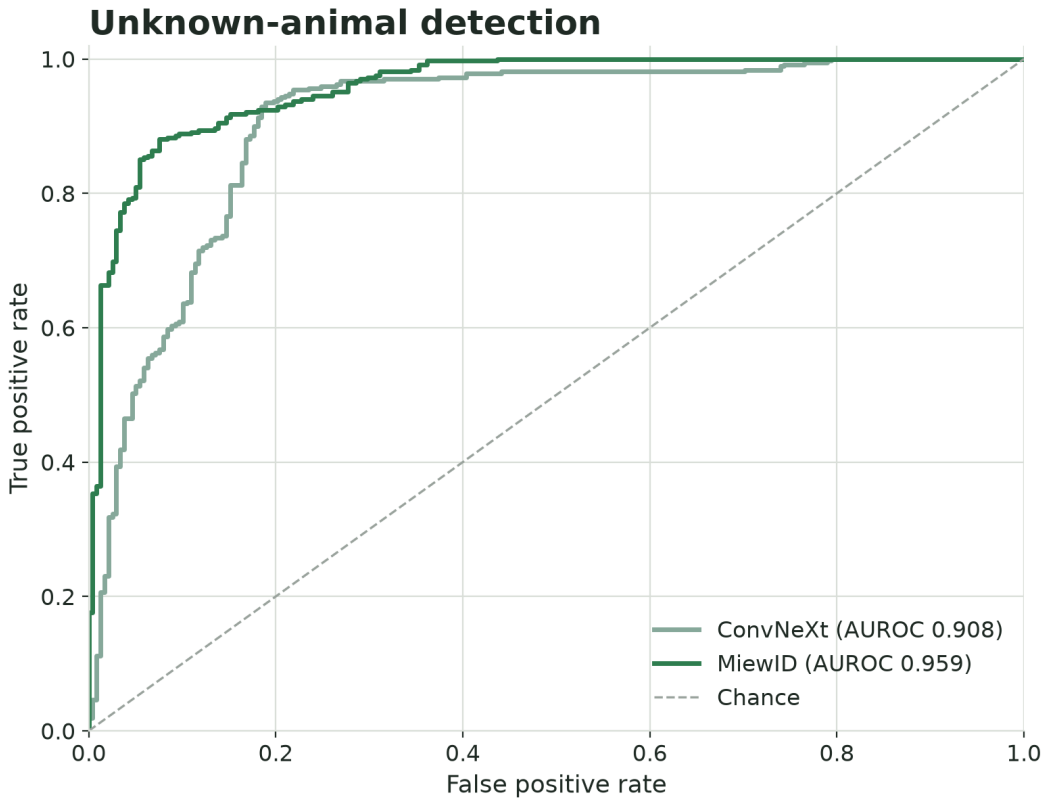


Figure 2. ROC curves treat a larger value of 1 minus the model's maximum score as stronger evidence that a query is unknown.

MiewID's AUROC of 0.959 indicates better score separation across possible operating thresholds. ConvNeXt also performs strongly at 0.908, but its curve remains below MiewID across most of the low false-positive region.

Efficiency and deployment

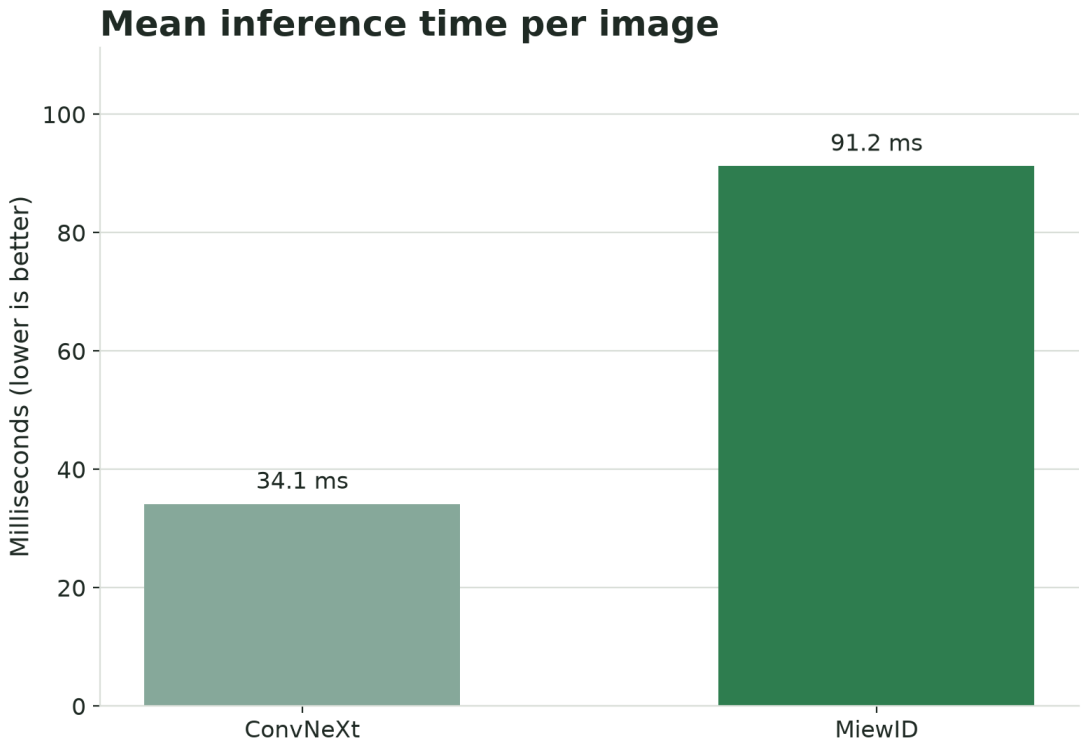


Figure 3. Mean per-image inference time on the same Apple Metal device. Model loading and image download time are excluded.

Deployment decision

MiewID is the recommended primary model because Top-1 and open-set AUROC best represent the application's goal: identify one known animal while recognizing when the animal is outside the known set. ConvNeXt remains a useful faster baseline and comparison option.

Limitations

The experiment compares complete pipelines rather than identical architectures. It covers a reproducible 100-identity subset, and the web application's UNKNOWN threshold has not been calibrated separately for each score type. Timing may also change on a CUDA GPU or production server.

Reproduce the comparison

Run `python src/models/evaluate_comparison.py --device auto`. The command verifies the shared identity list and writes both aggregate JSON metrics and a per-query CSV without retraining either model.